

LEC12 - Gravity & Tides

1/2

Learning Goals:

- Describe the phenomenon of tides on the Earth's surface
- Explain the causes of tides due to the differential gravity exerted on the Earth by the Moon and Sun
- Discuss the exchange of angular momentum caused by tides and its effects on the Earth's rotation and the Moon's orbit

Differential Gravitational Force: A gravitational force acting on an object such that portions of the object experience a stronger force

Recap of Lec 11:

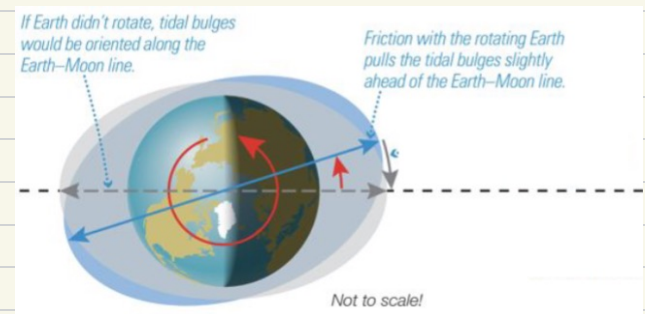
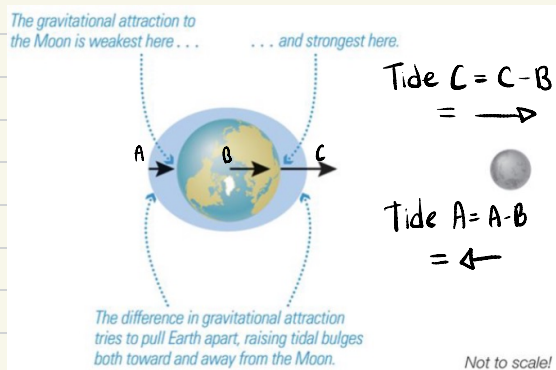
- A change in angular momentum caused by a net external torque

Tides:

- Differential gravitational force from the Moon stretches the Earth, raising symmetric tidal bulges
- Tides affect water and land, but since water flows more readily, we mostly observe changes in water level
- This model **does not** account for the complicated interactions between the oceans and continental boundaries, which make tides on Earth much more complicated than on an ocean planet

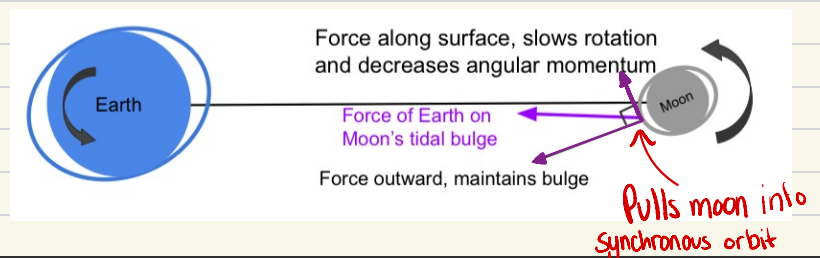
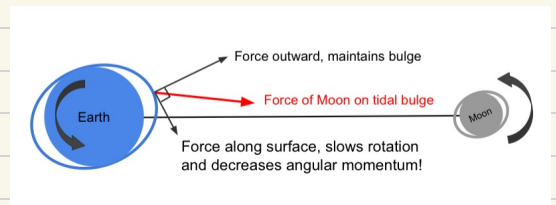
Why two tidal bulges:

- Differential gravitational force from the Moon (subtract the average force)
- This causes water to rush to the locations along the Earth-Moon axis
- Earth's rotation causes friction, dragging the tidal bulges along the Earth's surface in the direction of rotation



Tides and the Moon:

- The Moon also experiences tides due to Earth
 - Moon is smaller, so the difference in force between the near side and far side is smaller
 - No easily-moved water on the Moon
- The Moon's gravitational force on the Earth's tidal bulge acts as a torque to change the Earth's angular momentum
- In the past, the Moon rotated faster, causing a similar misalignment of its tidal bulges



- Tides are caused by the pull of the Sun and the Moon on the Earth. Earth has 2 distinct tidal bulges caused by differential gravitational forces
- The tides act as a torque force, which slow Earth's rotation speed (longer days & nights) as well as the Moon's until their tidal axis was aligned

LEC12 - Gravity & Tides

2/2

Synchronous Rotation:

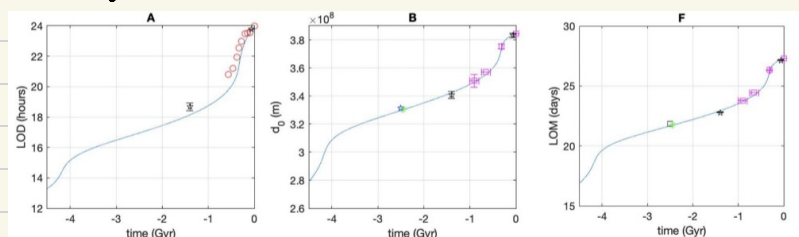
The orbital period = rotation rate (1 day = 1 year)

Milankovitch Cycles: First

covered in LEC6 -

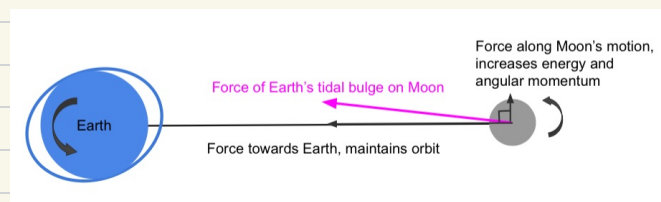
Constellations & Seasons

- Torque from the Earth's gravity slowed the Moon's rotation and locked it into **synchronous rotation**
- Once the Moon's rotation slowed enough that it was equivalent to its orbital period, the bulges are now perfectly aligned, so no further slowing occurs
 - Earth is still slowing down, but very slowly
- Earth and Moon used to rotate much faster
- The Moon used to be closer to Earth, so its orbital period was shorter
- Milankovitch cycles in banded iron formation constrain the Earth-Moon system 2.46 Billion years ago
 - Geologists demonstrate that the Moon was closer to Earth 2.5 billion years ago by studying how **Milankovitch cycles** have evolved since then



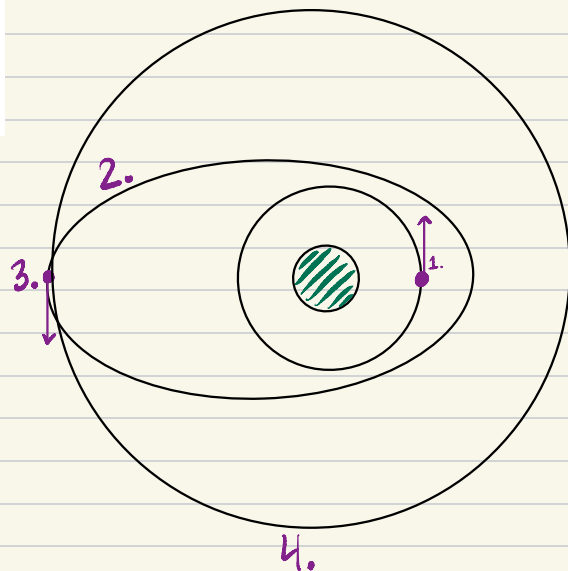
What happens to the angular momentum?

- The Moon's orbit increasing because it inherits the angular momentum lost by Earth's and Moon's rotation slowing down



Effect of Increasing Orbital Speeds:

1. Force boosts orbital speed
2. Orbit grows elliptical since speed is too high for circular orbit
3. Continued force boosts orbital speed more
4. This maintains circular orbit, but orbit grows gradually



Tidal Forces from the Sun:

- The Earth also experiences tides from the Sun
- The strength of the Sun's tidal effect on the Earth is about half that of the Moon

Spring & Neap Tides:

- Spring tides occur at full and new moons (forces add)
- Neap tides occur at first and third quarter moons (forces partially cancel)



- Milankovitch cycles are highly dependent on the gravity of the moon, and the tides
- The Sun also causes tides on Earth, but less intensely
- Spring tides occur when the Moon's and Sun's tides combine. Neap tides occur when they partially cancel.